

02 - IVPs; direction fields

1. **Daily Problem.** Last class we found that the functions $y(x) = Ce^x$ (for any real number C) are solutions of the ODE $\frac{dy}{dx} = y$. Now, find as many solutions as you can to the ODEs $\frac{dy}{dx} = 5y$ and $\frac{dy}{dx} = 5x$.



$$y = \frac{5}{2}x^2 + C$$

$$y = Ce^{5x}$$

A general solution of an ODE is

a representation of all solns of that ODE

Let's generalize a principle from the previous problem: the general solution of an ODE of the type

is given by:

An initial value problem (IVP) is

A problem in which

we must find a general soln for an ODE given ≥ 1 ICs

2. Solve the Initial Value Problem (IVP) $\frac{dy}{dx} = 5y$ with $y(-2) = 1$. How many solutions did you find?

general soln $y = Ce^{5x}$
 substitute in $x = -2$

$$1 = Ce^{-10} \quad C \text{ must be } e^{10}$$

exactly 1 soln, 1st degree ODE
 w/ 1 IC

3. Find the general solution to the differential equation $\frac{dy}{dt} = t^2 + 5$.

$$\frac{t^3}{3} + 5t + C$$

v.a Integration

4. (a) Solve the differential equation $\frac{d}{dt}(ty) = 5t$ for the unknown function $y(t)$.

- (b) Why does your strategy not work to solve $\frac{d}{dt}(ty) = 5y$ instead?

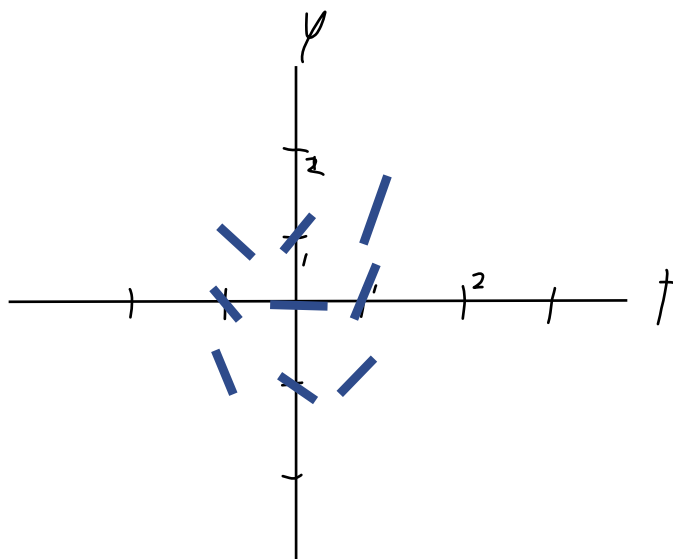
5. **Direction fields.** Consider the differential equation $\frac{dy}{dt} = 2t + y$. You can think of $\frac{dy}{dt}$ as depending on both t and y . For example, when $t = 1$ and $y = 3$ then $\frac{dy}{dt} = 2 \times 1 + 3 = 5$.

(a) ~~10 DE~~ Fill in the following table accordingly:

t	y	$dy/dt = 2t + y$
-1	-1	-3
0	-1	-1
1	-1	1
-1	0	-2
0	0	0
1	0	2
-1	1	-1
0	1	1
1	1	3

(b) How would you describe all the points (t, y) for which $\frac{dy}{dt} = 0$?

(c) Draw the direction field for this ODE.



Why construct a direction field?

To sketch a few representative solutions, and have a rough sense of what the solutions look like.

How to construct a direction field?

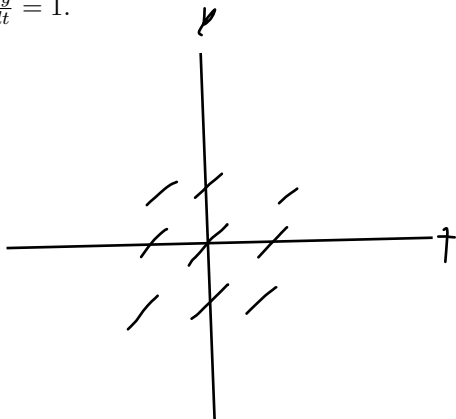
1. Draw a grid on the t, y plane.
2. At each grid point, draw short line segments that have slope dy/dt given by the ODE.

6. To simplify our work, and to give us better intuition for the solutions, we use isoclines. An isocline of an ODE $\frac{dy}{dt} = f(t,y)$ is a set of points (t,y) where the slope (cline) is always the same (iso). Which isocline did we consider in the last problem? (We call this special kind of an isocline a nullcline.)

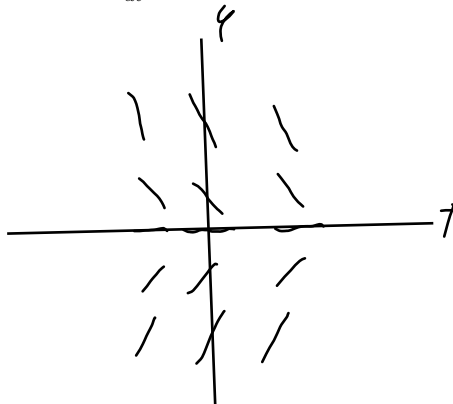
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7. Draw the direction fields for the following differential equations, with a few representative solutions.

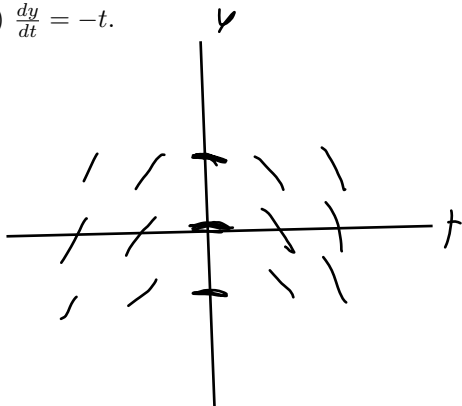
(a) $\frac{dy}{dt} = 1.$



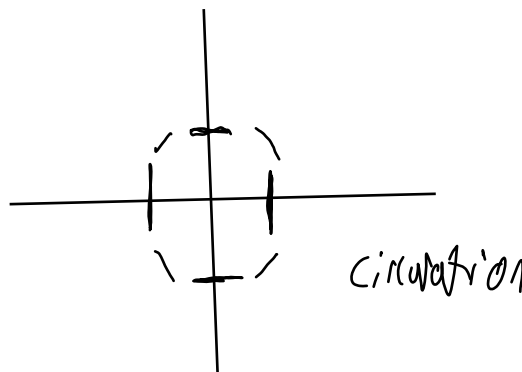
(b) $\frac{dy}{dt} = -y.$



(c) $\frac{dy}{dt} = -t.$



(d) **Daily Problem.** Please try this question in preparation for next class. $\frac{dy}{dt} = -\frac{t}{y}.$



8. Let's reflect back on some shortcuts we learned from the previous problem:

(a) If you have a differential equation of the form $\frac{dy}{dt} = f(t)$ (that is, $\frac{dy}{dt}$ depends only on t , not on y ; an example is $\frac{dy}{dt} = t^2 + 1$), what can you say about its direction field and isoclines? How do different solutions of the differential equation relate to each other?

It will be constant

(b) If you have a differential equation of the form $\frac{dy}{dt} = f(y)$ (that is, $\frac{dy}{dt}$ depends only on y , not on t), what can you say about its direction field and isoclines? How do different solutions of the differential equation relate to each other?

constant w/ t

9. For each direction field in #7, can you guess the general solution from looking at the direction field? Check your answer by plugging back in the ODE! (In a week or two, you should be able to solve each ODE analytically, so come back and do that, and make sure you get the same answers!)

a
f

b
 ce^{-t}

4

c
 $-\frac{1}{2}t^2$

d
↑

10. Here are four differential equations. Match each with a direction field below.

(a) $\frac{dS}{dt} = 2 - S/50$. (b) $\frac{dS}{dt} = 50(\frac{100}{S} - 1)$. ~~(c) $\frac{dS}{dt} = 2 - \frac{2S}{100-t}$~~ . ~~(d) $\frac{dS}{dt} = \frac{200-2S}{100-t}$~~ .

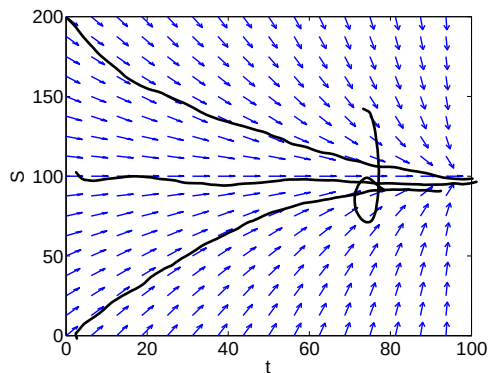


Figure 1: (A)

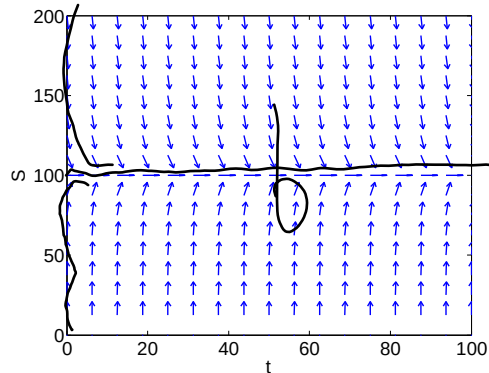


Figure 3: (C)

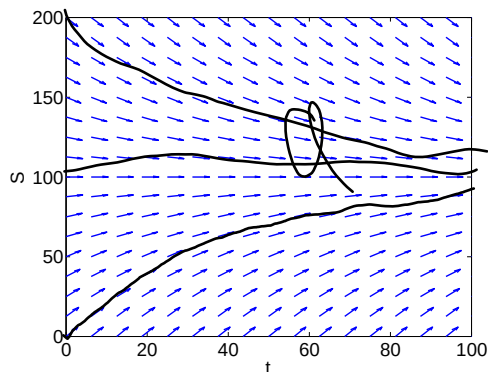


Figure 2: (B)

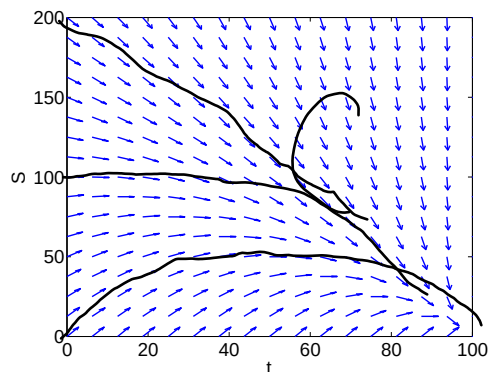


Figure 4: (D)

11. For each of the direction fields from the previous question, draw qualitatively correct solutions for the IVPs with $S(0) = 0$, $S(0) = 100$ and $S(0) = 200$. For each direction field, say whether the behavior of solutions as t becomes very large depends on the initial condition.

12. (a) Draw the direction field for the differential equation $\frac{dy}{dt} = 1 - y$. Sketch two solutions.

(b) Which of the following is a solution to $\frac{dy}{dt} = 1 - y$? (Can you guess from the direction field before trying them all out?)

i. $y = Ce^{-t}$
 ii. $y = Ce^t - t$

iii. $y = Ce^{-t} - 1$
 iv. $y = Ce^{-t} + 1$

v. $y = Ce^t + 1$

